

Electronic Companion

Neural Differential Utility: Accepted Adaptive Service Control and Certified Value-Gradient Learning

Revision R10, September 21, 2026

This companion contains current acceptance, accounting, switching, and computational details. Every predecessor main-paper section moved here remains intact. The separate historical supplement retains the full earlier diffusion and continuous-time development and all earlier computational tables; cross-document references identify their location. No historical numerical evidence is pooled with the nonlinear multistage experiment.

EC.1 Accepted contracts and the value of information

The relevant static comparator is chosen before period zero, not fixed at an arbitrary tier. Let $\mathcal{P}_{\text{stat}}$ consist of constant-contract protocols, with stock still optimized using the observed regime. Every comparator pays the initial installation cost $\lambda(\theta - q_0)^2$. Let $\mathcal{P}_t$, $\mathcal{P}_{t,z}$, $\mathcal{P}_{t,q}$, and $\mathcal{P}_{t,z,q}$ denote deterministic contract rules using the displayed information. Physical stock can use the full observed state in each class. With a fixed initial contract,

$$\mathcal{P}_{\text{stat}} \subseteq \mathcal{P}_t \subseteq \mathcal{P}_{t,z} \subseteq \mathcal{P}_{t,z,q}. \quad (\text{EC.1})$$

The inherited-only class is not an additional information increment along this chain.

Proposition EC.1.1 (Inherited-only information and optimized static contracts). *For deterministic q_0 and deterministic protocols, the value of $\mathcal{P}_{t,q}$ equals the value of $\mathcal{P}_t$. For a finite stock menu and continuous tier, the optimal static contract can be found exactly by enumerating stock-envelope breakpoints and the stationary point of the objective on each intervening interval.*

Proof. If $\theta_t = f_t(q_t)$ and $q_{t+1} = \theta_t$, induction makes the entire tier path deterministic. A time-only schedule implements that path, and every schedule is an inherited-only rule. Independent private randomization is a mixture of deterministic paths and cannot improve the unconstrained maximum.

For the static calculation define $a_z = \sum_t \beta^t \Pr(z_t = z)$ and $H = \sum_z a_z$. Its objective is

$$W_{\text{stat}}(\theta) = \sum_z a_z \{r_z \theta + U_z(\theta)\} - mH\theta^2 - \lambda(\theta - q_0)^2. \quad (\text{EC.2})$$

Each U_z is a finite upper envelope of affine stock payoffs. Between all pairwise line crossings, the active stock choices are constant, so (EC.2) is a concave quadratic with coefficient $-(mH + \lambda)$. Its maximum on that interval is at its stationary point or an endpoint. Checking all such candidates and the endpoints 0, 1 is exhaustive, including ties and nonconcavity across intervals. One may instead first construct each stock upper envelope and enumerate only its surviving breakpoints; intersections never attaining that envelope are unnecessary. Both procedures return the same exhaustive candidate set of active intervals. $\square$

Public randomized accepted classes are treated separately in Proposition 6.2; the unconstrained mixture argument above is not sufficient for them. The equality in the proposition concerns the value from the specified initial state; it is not equality of the two feedback function spaces on unreachable histories. For $\mathcal{P}_{t,z}$ we use occupation variables and binary tier selectors shared across all inherited states. Optimizing an ordinary full-state Bellman equation and then deleting q from the policy label would not solve this restriction.

EC.1.1 A constrained operating protocol

The customer-accepted problem is

$$\max_{\pi} W^{\pi} \quad \text{subject to} \quad F^{\pi} \geq \gamma D, \quad U^{\pi} \geq \underline{U}, \quad C^{\pi} \leq \overline{C}, \quad (\text{EC.3})$$

where the optional final constraint rules out financing provider gains with a higher physical resource cost. For a finite tier menu let $x_t(z, q, S, \theta)$ be the undiscounted state–action occupancy. It satisfies $x \geq 0$ and

$$\sum_{S, \theta} x_0(z, q, S, \theta) = \mu_0(z, q), \quad \sum_{S, \theta} x_t(z, q, S, \theta) = \sum_{z-, q-, S-} P_{z-z} x_{t-1}(z-, q-, S-, q), \quad t \geq 1. \quad (\text{EC.4})$$

All objectives and commitments are linear functionals of x , with β^t in their coefficients. The transition on the right has next contract q , explaining the last action index. The full stock menu is retained: eliminating stock by the unconstrained newsvendor solution is generally invalid under service and participation constraints.

Theorem EC.1.2 (Acceptance, Pareto comparison, and a certificate). *For finite states and actions, (EC.3) equals the linear program defined by (EC.4) and its linear commitments. From any feasible occupation measure, choosing actions with probabilities proportional to the local occupancies implements the same reward and commitments. If a static comparator is feasible, the optimum in a policy class containing it is at least its reward. Moreover, for any $\mu, \eta, \chi \geq 0$,*

$$W_{\text{acc}}^* \leq V_0^{\mu, \eta, \chi} - \mu \gamma D - \eta \underline{U} + \chi \overline{C}, \quad (\text{EC.5})$$

where $V^{\mu, \eta, \chi}$ is the ordinary Bellman value for reward $g + \mu f + \eta u - \chi C$. A feasible protocol and this upper bound give a two-sided certificate without trusting an optimizer's status label.

Proof. History-dependent policies induce occupancies satisfying the flow equations. Conversely, normalize each positive local occupancy; induction on t recovers every occupancy, choosing arbitrary actions at zero-mass states. This proves equality and implementability. The finite feasible flow polytope is compact, and linear programming duality applies when the commitment set is nonempty. The comparison follows by feasibility. For the certificate, each feasible policy satisfies $W^{\pi} \leq W^{\pi} + \mu(F^{\pi} - \gamma D) + \eta(U^{\pi} - \underline{U}) - \chi(C^{\pi} - \overline{C})$. Maximize the expression over all policies and evaluate its reward term by Bellman recursion. $\square$

The same Lagrangian retains the inherited-contract envelope because the added commitments depend on current stock and tier, not separately on inherited q . However, the premium–liability monotonicity assumptions may change after reweighting. Neither a constrained optimizer nor its randomized policy is assumed to inherit the relaxation's statewise monotonicity without checking those conditions.

For the canonical primitives in Section EC.5, optimize the continuous static tier and set $(\gamma D, \underline{U}, \bar{C})$ equal to its fill, customer utility, and physical cost. The exact stationary candidate is

On the 0.1 tier grid, a dynamic protocol achieves

with exactly equal F , U , and C to the continuous static comparator. The finer static class makes this comparison more demanding than comparison with the best static grid point. Randomizing over static contracts cannot improve their unconstrained maximum; the static optimum itself satisfies these commitments.

This statement has a constructive rational certificate. Every positive-occupancy stock is $S = z + 2$. The protocol is deterministic except at $(t, z, q) = (1, 2, 0.5)$, where it selects tier 0.8 with probability

and tier 0.7 otherwise. The complete feedback map is supplied with the certificate. Fixing that single choice low or high gives two deterministic protocols with rational rewards and utilities. Set $\omega = (\underline{U} - U_L)/(U_H - U_L)$; exact forward propagation verifies $0 \leq \omega \leq 1$ and equality of all three commitments. The gain in (EC.6) is the positive rational number

An independent rational Bellman recursion for (EC.5), using nonnegative rational prices and $\chi = 0$, gives an upper bound less than 8.21×10^{-16} above this feasible reward. Thus the observed improvement is neither Monte Carlo noise nor a payment for lower service. It is an instance-specific, reproducible Pareto improvement in provider reward at the customer's reservation utility and service level; it is not a universal dominance theorem for every set of primitives.

For a policy π , let C^π be its expected discounted physical cost and let K^π be its expected discounted premium minus liability, maintenance, and adjustment. Its net reward is $W^\pi = K^\pi - C^\pi$. All these quantities use the same external coefficients. Let J denote a net-reward-optimal joint policy, F a frozen-contract policy optimized for that reward, and O a physical-cost-optimal policy with the same information and admissible stock actions.

Theorem EC.2.1 (Accounting and an optimizer-based physical-cost bound). *For these actual policies,*

$$C^F - C^J = (W^J - W^F) - (K^J - K^F), \quad (\text{EC.7})$$

$$0 \leq C^J - C^O \leq K^J - K^O. \quad (\text{EC.8})$$

Thus J improves physical cost over F exactly when its optimized net-reward gain exceeds the increase in net contract transfers. Physical-cost dominance over O cannot follow from net-reward optimality. A physical oracle may have multiple contract implementations; (EC.8) holds for any specified feasible implementation.

Proof. Subtract $W = K - C$ for the two policies to obtain (EC.7). Physical optimality gives $C^O \leq C^J$. Net-reward optimality gives $K^J - C^J \geq K^O - C^O$, yielding the other inequality. Neither step inserts a preferred contract into the optimizing policy. $\square$

The bound can be strengthened by choosing the net-transfer-maximizing implementation within the physical-optimal policy set. Our primary experiment does not change the previously recorded oracle implementation for this purpose: the greatest-contract tie convention selects contract one. The physical lower bound is independent of that convention, whereas the oracle's reported net cash flow is not.

A positive physical gap can also be stable without assuming that all reoptimized policies remain unchanged. Suppose the same two calibrated policies are deployed under a perturbed exogenous one-step law whose total-variation distance is at most η uniformly in regime, and physical stage cost lies in $[0, \bar{C}]$. Coupling exogenous histories gives

$$|(C^F - C^J)_{\text{new}} - (C^F - C^J)_{\text{old}}| \leq 2\bar{C}\eta \sum_{t=0}^{T-1} \beta^t (t+1). \quad (\text{EC.9})$$

Before a coupling failure the two copies of each fixed policy choose identical contracts and stock. The chance of failure by period t is at most $(t+1)\eta$; summation proves the bound. This statement concerns deployment of the calibrated policies, not arbitrary policy reselection at a tie. The traditional critical-fractile mismatch argument is retained separately in EC.2; it is not used to attribute physical improvement to an unproved endogenous selector.

EC.2.1 Switching thresholds and persistence

Let

$$B_t(z, \theta) = r_z \theta - m\theta^2 + U_z(\theta) + \beta \sum_{z'} P_{zz'} V_{t+1}(z', \theta).$$

This notation removes the current adjustment term, but its continuation still depends on the adjustment-cost parameter. Write $b_i = B_t(z, \theta_i) - \lambda\theta_i^2$ for a finite ordered menu.

Proposition EC.2.2 (Thresholds and two-tier hysteresis). *For $\lambda > 0$, adjacent surviving lines $i < j$ of the upper hull exchange optimality at*

$$q_{ij} = \frac{b_i - b_j}{2\lambda(\theta_j - \theta_i)}. \quad (\text{EC.10})$$

The greatest optimizer selects the higher tier at equality. Suppose the menu consists of two tiers $a < b$ and put $\Delta = B_t(z, b) - B_t(z, a)$ and $d = b - a$. From inherited tier a the provider switches upward exactly when $\Delta \geq \lambda d^2$. From inherited tier b the provider switches downward exactly when $\Delta < -\lambda d^2$. Thus the interval $-\lambda d^2 \leq \Delta < \lambda d^2$ preserves the inherited tier.

Proof. Subtract the two affine maximands in (13) and solve their equality for q . Ordered slopes and deletion of dominated lines give ordered crossings. With two tiers, the upward value difference at $q = a$ is $\Delta - \lambda d^2$; at $q = b$ it is $\Delta + \lambda d^2$. The strict and weak inequalities follow from selecting the greater tier at a tie. $\square$

The hysteresis width is $2\lambda d^2$ in continuation-adjusted reward units. This is an implementable rule: change tier only when its incremental premium, liability, maintenance, and continuation advantage crosses the appropriate adjustment threshold. It is not a claim that a threshold expressed in an arbitrary economic signal grows monotonically with λ : B_t changes with λ too. For a continuous menu, a quadratic adjustment cost generally gives small continuous adjustments rather than a positive inaction band. For a larger discrete menu, a pairwise comparison is a decision threshold only when both lines survive the hull. These distinctions connect the present structure to, rather than replace, the hysteretic-control literature (Hipp and Holzbaur, 1988).

Proposition EC.2.3 (Aggregate comparative statics). *Fix the transition law, initial state, discount, horizon, and a parameter-independent feasible policy class. If a stage coefficient a enters reward as $-ae_t$, then any optimizers π_1, π_2 at $a_1 < a_2$ satisfy*

$$\mathbb{E}^{\pi_2} \sum_t \beta^t e_t \leq \mathbb{E}^{\pi_1} \sum_t \beta^t e_t.$$

The optimal value is convex and nonincreasing in a when $e_t \geq 0$. Consequently optimal discounted squared movement is nonincreasing in λ , tier square in m , and indemnity exposure $\theta L_z(S)$ in κ . A uniform additive premium increase makes optimal discounted tier usage nondecreasing.

Proof. Write a policy's objective as $A(\pi) - aE(\pi)$. Optimality at the two coefficients gives $A_1 - a_1 E_1 \geq A_2 - a_1 E_2$ and $A_2 - a_2 E_2 \geq A_1 - a_2 E_1$. Adding yields $(a_2 - a_1)(E_1 - E_2) \geq 0$. A supremum of affine functions is convex; their nonpositive slopes give monotonicity. Replace $-aE$ by $+aE$ for the premium statement. $\square$

These are policy-level exposure results, not pointwise monotonicity of each action. In particular, the κ comparison does not hold automatically when changing κ also changes the participation-feasible set through (3). Neither discounting nor regime persistence is a linear coefficient of a fixed policy's discounted occupation measure; their effects are reported by recomputation, without a universal sign assertion.

A constant contract can be optimal even with state variation. If one tier $\bar{\theta}$ maximizes $r_z \theta - m\theta^2 + U_z(\theta)$ for every regime and $q_0 = \bar{\theta}$, selecting it forever attains every period's intrinsic upper bound with zero adjustment. It is therefore optimal for every horizon, discount, transition law, and $\lambda \geq 0$. Conversely, premium-liability compatibility is substantive. For $T = 1$, $S = \{0\}$, deterministic $D_z = z$, $r_z = 1$, $q = 1/2$,

$m = 3/2$, and $\lambda = 3/5$, the continuous optimizer is

$$\theta^*(z) = \text{proj}_{[0,1]} \frac{1.6 - 1.2z}{4.2}.$$

It decreases from $8/21$ to $2/21$ and then to zero as $z = 0, 1, 2$. This example has a positive external premium; failure is not confined to a self-selected penalty with no counterparty. Compatibility is sufficient, not necessary: a particular calibrated family may remain monotone after the sufficient inequality fails.

EC.2.2 Beyond scalar quadratic adjustment

For a continuously differentiable strictly convex potential ϕ on a compact contract interval, define

$$D_\phi(\theta, q) = \phi(\theta) - \phi(q) - \phi'(q)(\theta - q).$$

Replace $\lambda(\theta - q)^2$ by $\lambda D_\phi(\theta, q)$; $\phi(q) = q^2$ recovers the original model. The cost can be asymmetric. This is a specified nonquadratic cost family, not a representation of every possible switching cost.

Theorem EC.2.4 (Dual-coordinate envelope). *Suppose current q enters only the adjustment cost and the next inherited state equals θ . Other state transitions and rewards may depend on an operating state x , regime, stock, and θ , but not separately on q . Then*

$$V_t(x, z, q) = \lambda\{\phi(q) - q\phi'(q)\} + \max_{\theta} \{b_t(x, z, \theta) + \lambda\theta\phi'(q)\}, \quad (\text{EC.11})$$

where b_t includes the optimization over physical actions and all continuation terms, and subtracts $\lambda\phi(\theta)$. For a scalar ordered finite menu, the same linear-time hull construction and ordered queries apply in the coordinate $y = \phi'(q)$. In dimension d , the identity holds with $q\phi'(q)$ replaced by $q^\top \nabla \phi(q)$ and with an envelope of affine functions of $y = \nabla \phi(q)$.

Proof. Expand $-\lambda D_\phi$ in the Bellman objective. Terms depending solely on q leave the maximum; all remaining dependence on q is $\lambda\theta\phi'(q)$. Convexity makes ϕ' nondecreasing. The scalar affine slopes $\lambda\theta$ are ordered, so the hull proof of Theorem 3.3 applies verbatim to ordered dual coordinates. The vector identity follows from the same expansion. $\square$

In dimension greater than one the cells are polyhedral in dual coordinates; the identity does not give a linear-time multidimensional hull algorithm. A continuous physical state changes the intercept computation and may still require numerical integration and physical-state discretization. The theorem accelerates the inherited-contract dimension, not every dimension of a general inventory system.

Nonlinear menu payments need not destroy the structure. If premium and liability are $r_z h(\theta)$ and $\kappa h(\theta) L_z(S)$ for a continuous nondecreasing h , the stock–tier and regime–tier cross differences retain their signs under Assumption 3.1; arbitrary continuous tier-only maintenance does not change those cross differences. With quadratic adjustment, Theorem 3.2 still applies. The envelope identity also persists because it uses the adjustment term, not linearity of tier payments. However, the particular convexity argument giving

the $O(n^{-2})$ bound must be checked anew when h changes. We do not transfer that rate to arbitrary nonlinear payments by assertion.

EC.2.3 The actual multi-tier inaction regions

A pairwise crossing alone need not be a surviving decision region. The following statement compares every competing tier. Fix the continuation value and let a one-review premium shift s add $s\theta$ to its intrinsic reward B_i ; future premiums are not changed. For inherited tier θ_j , define

$$\ell_j = \max_{i < j} \left\{ \frac{B_i - B_j}{\theta_j - \theta_i} - \lambda(\theta_j - \theta_i) \right\},$$

$$u_j = \min_{i > j} \left\{ \frac{B_j - B_i}{\theta_i - \theta_j} + \lambda(\theta_i - \theta_j) \right\},$$

with the empty maximum $-\infty$ and empty minimum $+\infty$.

Proposition EC.2.5 (Surviving inaction bands). *Under the greatest-tier tie convention, the installed tier θ_j is retained exactly when $\ell_j \leq s < u_j$. An empty interval means that this tier has no inaction band. For a uniform menu of width h and concave intrinsic reward sequence, the finite endpoints reduce to*

$$\ell_j = -\frac{B_j - B_{j-1}}{h} - \lambda h, \quad u_j = -\frac{B_{j+1} - B_j}{h} + \lambda h.$$

Both endpoint sequences are nondecreasing. Each interior band's width is $[(B_j - B_{j-1}) - (B_{j+1} - B_j)]/h + 2\lambda h$, and adjacent bands overlap by exactly $2\lambda h$.

Proof. Subtract the payoff from every lower and higher competing tier at inherited state $q = \theta_j$. Every lower comparison imposes the corresponding lower bound, whereas a tie with a higher tier excludes the upper endpoint. Their intersection is precisely the displayed interval. Under concavity, the largest lower restriction and smallest upper restriction come from the adjacent tiers. Ordered adjacent secant slopes give ordered endpoints and the stated width and overlap formulas. $\square$

Concavity is a sufficient condition for the simplified formulas, not an implication of the general inventory envelope. Our diagnostic therefore uses all competitors for each of the 264 period–regime–tier cases and stores empty bands as well as surviving ones. A choice cell in the inherited coordinate is not called an inaction interval when the chosen tier differs from that coordinate. Neither this result nor a fixed-continuation shift is used as a permanent-premium comparative-static theorem.

EC.3 Retained constructive smooth neural critics

The exact finite-grid envelope also supplies a smooth neural approximation with a deterministic error budget. This construction uses computed envelope weights rather than statistical training. It provides a certified neural representation of the contract model without making a claim about the optimization of a large network.

Corollary EC.3.1 (Smooth value-gradient representation). *For a finite contract grid, let $V_t^{(n)}$ be the exact grid value, extended to every inherited $q \in [0, 1]$ by (13). For each (t, z) write its convex part as*

$$V_t^{(n)}(z, q) + \lambda q^2 = a + bq + \sum_j d_j(q - k_j)^+, \quad d_j \geq 0, \quad \sum_j d_j \leq 2\lambda.$$

For $\tau > 0$, replace each hinge by

$$h_\tau(x) = \frac{(x + \tau)_+^3 - 2x_+^3 + (x - \tau)_+^3}{6\tau^2}$$

and denote the resulting critic by $w_{t,\tau}$; keep $w_{T,\tau} = 0$. It is a quadratic skip connection plus a finite cubic-ReLU network, is C^2 in q , and satisfies

$$0 \leq w_{t,\tau}(z, q) - V_t^{(n)}(z, q) \leq \lambda\tau/3. \quad (\text{EC.12})$$

If A_τ exactly maximizes the one-step grid objective with continuation $w_{t+1,\tau}$, then, for $r = T - t$,

$$0 \leq V_t^{(n)}(z, q) - V_t^{A_\tau}(z, q) \leq \beta(\lambda\tau/3)H_{r-1}. \quad (\text{EC.13})$$

Combining this bound with (14) bounds loss relative to continuous contract actions. The derivative $\partial_q w_{t,\tau}$ is a compatible shadow price, not an independently fitted vector field.

Proof. The hinge representation follows from the ordered slopes of the upper envelope. Its total slope increase is at most 2λ , the width of the slope interval. The cubic finite difference equals the convolution of x^+ with the symmetric triangular density $(\tau - |y|)_+/\tau^2$. Convexity gives $h_\tau(x) \geq x^+$, and direct integration shows $h_\tau(x) - x^+ \leq \tau/6$, with equality at $x = 0$. This proves (EC.12). Each cubic-ReLU term is C^2 .

The exact and approximate one-step action values differ by a number in $[0, \beta\lambda\tau/3]$. Thus maximizing the latter loses at most $\beta\lambda\tau/3$ in the former, rather than twice that amount, because the approximation is one-sided. The final action uses the exact terminal continuation and has zero loss. Backward policy evaluation gives (EC.13). The gradient statement follows from the scalar construction. $\square$

Network size convention. Each hidden unit computes $(wq + b)_+^3$; three units implement one smoothed hinge. The reported maximum of 27 units is per (t, z) critic, excluding its explicit quadratic skip $-\lambda q^2$ and affine skip. There is no sharing across the 24 nonterminal (t, z) critics; their aggregate hidden-unit budget is at most $24 \cdot 27 = 648$, not 27 for one joint network. Each unit has a scalar input weight, bias, and output weight; skip coefficients must also be stored. This convention is not directly comparable to the 1,281 trainable parameters of the joint tanh critic in Section H.4.

EC.4 Continuation participation and its rational certificate

Index the regime-history decision nodes by i , with discounted probability $w_i > 0$, parent $p(i)$, and tier x_i . For the root, the parent tier is the fixed q_0 . Let a_i denote its net payment rate. The interim constraints can be

multiplied by the unconditional discounted probability of their root without changing their meaning. Their matrix form is

$$Cx \leq h, \quad C_{ij} = w_j a_j \mathbf{1}_{\{j \text{ descends from } i\}}, \quad h_i = \bar{\theta} \sum_{j \text{ descends from } i} w_j a_j.$$

Write the reward as $W(x) = -x^\top Hx + b^\top x - c_0$. The adjustment graph contributes nonnegative squares, so $H \succeq m \text{diag}(w)$. The change of variables $y_i = \sqrt{w_i}x_i$ gives a well-conditioned sparse quadratic matrix for the numerical proposal; a sparse factorization is reused in the bound-constrained dual solve. There are only $O(TN)$ nonzero continuation coefficients on an N -node finite tree, rather than a dense N^2 constraint matrix.

Proposition EC.4.1 (Weighted residual certificate). *For any feasible $x \in [0, 1]^N$, nonnegative vector ℓ , and residual $r = b - 2Hx - C^\top \ell$,*

$$W^{\text{interim},*} - W(x) \leq \ell^\top (h - Cx) + \sum_i \frac{r_i^2}{4mw_i}.$$

All terms can be computed exactly from rational node probabilities, tier choices, and multipliers, even when a floating-point optimizer proposed them.

Proof. The Lagrangian $W(y) + \ell^\top (h - Cy)$ is an upper bound on every feasible reward. Its difference from its value at x is $r^\top (y - x) - (y - x)^\top H(y - x)$. Bound H below by $m \text{diag}(w)$ and complete each scalar square. Adding the slack term proves the result. Optimizing over all real vectors only relaxes the tier box. $\square$

The numerical proposal is rounded down to denominator 10^9 . If a continuation budget is violated after rounding, multiply all tiers by the smallest ratio of the corresponding budget to its proposed subtree payment, then round down again. Since every a_i is positive, this enforces every subtree constraint and preserves the tier box. In the recorded calculation the contraction is approximately 0.99999943, not a material change in the policy. The proposed nonnegative dual variables are independently rounded; no stationarity claim is inferred from solver success.

The resulting deterministic reward is -5.4170322098230885 , and the exact upper-minus-lower gap is $6.948160845151524 \times 10^{-7}$. The tier range is approximately $[0.087564039, 0.672234204]$. All 3,280 continuation inequalities, the customer root constraint, physical cost, and service are satisfied. The displayed gap includes both weighted stationarity error and complementary slackness, and does not drop the larger of the two. The JSON certificate contains every rational tier and multiplier in breadth-first tree order. A separate verifier reconstructs the tree and repeats the audit without the numerical optimizer.

EC.5 Computational study of accepted adaptivity

We use the same canonical operating primitives throughout the exact comparison: $T = 8$, $\beta = 0.97$, $q_0 = 0.5$, $z_0 = 1$, and $D_z = z + D_0$, where $D_0 = 0, 1, 2, 3$ has probabilities 0.1, 0.3, 0.4, 0.2. The regime transition

matrix is

$$P = \begin{pmatrix} .70 & .25 & .05 \\ .15 & .70 & .15 \\ .05 & .25 & .70 \end{pmatrix}.$$

Stock is initially $0, 0.5, \dots, 6$ and the historical tier menu is $0, 0.1, \dots, 1$. Coefficients are $c = 0.3$, $h = 0.6$, $p = 0.65$, $\kappa = 1.2$, $m = 1.5$, $\lambda = 0.6$, and $r_z = 0.6 + 1.4z$. Customer service valuation is $v = 2$. A common retainer of 6 can be included on both sides of every provider comparison; the tabulated rewards exclude that constant. The continuous accepted hierarchy permits both stock and tier intervals, using the stock-identification proof rather than an unverified interpolation of the menu optimum.

The initial finite model has 264 nonterminal state nodes and 37,752 stock–tier choices over those nodes. It is small enough for an exact certificate, not presented as a demanding neural benchmark. Its function is to identify economic mechanisms and provide an independently checkable reference. The coupled portfolio experiment has a different function: it tests the accuracy, acceptance, and repeated-solve cost of a learned representation at larger dimension.

EC.5.1 Accepted information and menu resolution

Table 1 gives the accepted information decomposition. The time increment is 0.000816876783, the subsequent regime increment is 0.616558016063, and the inherited-tier increment is 0.019290378765. These four optimizations have identical customer, fill, and physical-cost reservations. The continuous full-state optimum has provider reward $-5.233445701435 \dots$ and exactly the static customer’s utility and service. Its exact coefficients are available for direct policy deployment without a neural network.

The older unconstrained information comparison is a different experiment: its inherited-tier increment is 0.050593409253. We retain that result in the companion but do not substitute it for the accepted increment. Likewise, the historical frozen tier of 0.5 is not used as the principal accepted benchmark. The continuous static optimum improves its reward by 0.076946642417 before any adaptivity is introduced.

Table 1 checks whether coarse tiers and randomization generate the accepted benefit. The randomized optimum on 320 intervals is within approximately 1.10×10^{-5} of the exact continuous optimum. A deterministic incumbent on that menu gains 0.63663320 over the continuous static contract. The best continuous protocol needs no lottery at all. The extracted deterministic incumbents are not claimed to be globally optimal on their discrete menus; one less favorable extraction at 20 intervals is retained rather than replaced by an invented monotone sequence.

EC.5.2 Robustness under the actual acceptance constraints

The new sensitivity study solves the accepted full-stock occupation program, not only a provider Bellman relaxation. Each perturbed primitive has a newly optimized continuous static reference. Its fill, customer utility, and physical cost set the reservations unless that row explicitly changes a reservation. When customer utility is raised, we also recompute the accepted static tier before reporting its gain. A comparison with an infeasible old reference would not establish accepted dominance.

Table 1: Accepted fine-menu optima and deterministic incumbents

Tier intervals	Randomized optimum	Deterministic incumbent	Incumbent gap
10	-5.244970661	-5.247006823	0.002036162
20	-5.237637537	-5.284466721	0.046829184
40	-5.234637362	-5.236634206	0.001996845
80	-5.233738433	-5.236860106	0.003121673
160	-5.233499396	-5.233703759	0.000204363
320	-5.233456635	-5.233477772	0.000021137

Same customer, fill, and cost reservations as Table 1. Only the first randomized row also has an independent rational R6 certificate. Other menu bounds are floating-point Lagrangian bounds. Deterministic entries are feasible incumbents, not claimed deterministic optima; their nonmonotone gaps reflect extraction, not nonnested feasible sets. The exact continuous optimum is $-5.233445701435 \dots$

The 31-case design varies adjustment cost, maintenance curvature and the maintenance power, the indemnity coefficient, premium intercept and slope, customer service valuation, regime persistence, demand dispersion, demand probabilities, utility and fill reservations, and the stock grid. Tier resolution is varied separately in Table 1. For every feasible LP we record the primal reward, flow residual, constraint slack, reduced-cost residual, and primal–dual gap. These floating-point records are numerical evidence, not rational certificates for every perturbation. Continuous information decompositions are reported only when the stock-support and interior quadratic checks succeed.

Table 2 displays selected cases; EC Table 3 contains every row. Of 31 accepted programs, 29 are feasible; 28 of those exhibit a positive gain over the applicable continuous accepted static comparator. A zero regime-premium slope removes the accepted gain in the reported instance. Increasing the fill floor by 0.015 is infeasible under the original physical-cost cap, and remains infeasible with the tested cap relaxation of 0.6. Both adverse rows are retained. Thus the conclusion is neither universal feasibility nor positive gain for arbitrary service promises.

Table 2: Accepted-problem sensitivity: selected cases

Case	Grid gain	Active	Lottery	Memory gain
canonical	0.625140	FUC	1	0.019290
lam = 0.1	0.728444	FUC	1	0.001115
lam = 4.0	0.361126	FUC	1	0.089863
power = 1.5	0.983858	FUC	1	—
power = 3.0	0.377330	FUC	1	—
persistence = 0.6	0.580915	FUC	1	0.022693
dispersion = 1.2	0.603774	FUC	1	0.018567
utility delta = 0.2	0.585697	FUC	1	0.018221
fill delta = 0.015	infeasible	—	—	—
zero regime premium	0.000000	FUC	0	—

Definitions and all 31 cases appear in EC Table 3. Reservations and static tiers are recomputed within each perturbed problem. The higher-fill case is infeasible under its stated physical-cost cap. The zero-slope case has no accepted gain.

Increasing the maintenance power from two to three leaves a positive accepted grid gain of approximately 0.37733; power 1.5 yields approximately 0.98386. Altering dispersion about the same mean also leaves positive gains in both tested directions. These results complement, rather than prove, Theorem 6.4. The theorem identifies a customer-neutral direction under explicit strict inequalities, whereas the numerical study allows stock changes and solves the full constrained problem. Under stock identification, changing ν alone cancels between the policy and the reservation because filled demand is the same; that invariance is an accounting implication, not new empirical evidence.

The inherited-state increment changes with the commitments and primitives. For example, it is about 0.00111 when $\lambda = 0.1$ and 0.08986 when $\lambda = 4$, even though total accepted grid gain falls in this slice. A cost can reduce the overall benefit of adaptation while making information about the installed tier more valuable. This distinction is obscured by reporting only static-versus-dynamic gains.

EC.5.3 Commitment and implementation

The customer-exit program imposes 3,280 continuation constraints on the original eight-review tree. The reported rational protocol satisfies every one, has reward $-5.417032209823\dots$, and retains a gain of $0.453078763224\dots$. Its upper-bound gap is below 6.95×10^{-7} . This result shows directly what survives when customer commitment is relaxed; it does not assume that an ex ante participation check enforces participation later.

For the coupled portfolio study, all 288 submitted decisions are stored as rational tier vectors, together with their nonnegative payment multipliers, graph matrices, customer slack, and loss bounds. An independent checker uses Python integer sparse multiplication, verifies the graph coercivity condition, and recomputes every fraction without calling an optimizer. At 1,024 services, the maximum learned value–gradient loss bound is 8.0436×10^{-7} per service, compared with a minimum realized gain of approximately 0.016067 per service over the optimized uniform tier. This is a decision-relevant bound on the accepted update, not the earlier finite-chain global residual budget.

The learned fit has a modest average loss advantage over its equal-oracle value-only counterpart, but the paired seed intervals contain zero. Conjugate gradients have substantially better accuracy in this experiment. The learned critic nevertheless supplies a compact compatible representation, accepted decisions, transfer across graph sizes, and a measured deployment tradeoff relative to refactorization. The result is a reproducible method comparison, not evidence that every numerical control problem requires a neural method.

EC.6 Operational implications

An accepted tier revision reallocates payments and maintenance resources; it is not merely a provider’s change of performance metric. Three ingredients make its value interpretable. The static alternative must be optimized. Customer service and participation must be enforced in the same comparison. Information classes must be optimized separately under those commitments. With those checks, the canonical gain remains positive, its sources can be separated, and costly inherited-state information has a measurable incremental value.

The exact scalar solution, the interim-participation extension, and the learned portfolio update answer successive implementation questions. They use different computational representations but the same acceptance logic. Complete contractibility remains an assumption, private-information incentive constraints remain distinct, and the portfolio benchmark is a coupled accepted review rather than a solution of an unrestricted high-dimensional multistage problem. The retained continuous-rate and diffusion analyses are labeled as extensions, with their boundary and discretization assumptions intact.

Reproducibility. Separate entry-point scripts implement exact information values, robustness, interim participation, portfolio learning, inaction, and independent verification in `revisions/or-r7-20260921/`. The scripts record rational certificates separately from floating-point sensitivity results. Exact quantities are compared as fractions; tolerances are used only for the explicitly numerical programs. Historical source directories remain unchanged. The response document maps every referee concern to the current result and records the remaining scope boundaries.

EC.7 Complete accepted sensitivity and learning diagnostics

EC.7.1 Perturbations, reservations, and information restrictions

Table 3 reports the entire 31-case design. Unless specified otherwise, one primitive changes and all others remain canonical. For maintenance power p , cost is $1.5\theta^p$. A persistence value ρ_P replaces the transition by $\rho_P P + (1 - \rho_P)\mathbf{1}(.25, .5, .25)$; both tested matrices remain stochastic. Dispersion rescales the four demand support points about mean 1.7, while the two probability perturbations keep the same support and mean with probabilities $(.05, .35, .45, .15)$ and $(.15, .25, .35, .25)$. Utility changes are absolute discounted-utility increments. Fill changes are increments in the ratio of discounted filled demand to discounted demand. The stock menu runs from zero to six at its specified step.

The continuous static comparator is reoptimized at every primitive. A changed reservation can make that original static tier infeasible; the JSON contains an explicit feasibility flag. For tighter utility reservations with an identified physical policy, the accepted static quadratic program is solved under the new budget, and that feasible value is used in the displayed gain. No positive gap to an infeasible old baseline is called Pareto improvement.

The full stock-tier LP is used in every sensitivity row. Stock is not removed by inserting the provider-relaxation newsvendor solution. A supporting-line search and endpoint checks determine whether the continuous quadratic information hierarchy is also available. Nonquadratic maintenance, slackened physical commitments, or a boundary optimizer may prevent that interior reduction; the continuous decomposition is then left unreported rather than inferred from the unconstrained problem.

Table 3: Full accepted-problem robustness design

Perturbation	Grid gain	Active	Lottery	Memory gain
canonical	0.625140	FUC	1	0.019290
lam = 0.1	0.728444	FUC	1	0.001115
lam = 1.2	0.549602	FUC	1	0.042054
lam = 4.0	0.361126	FUC	1	0.089863
m = 1.2	0.763126	FUC	1	—
m = 1.8	0.535059	FUC	1	0.012662
power = 1.5	0.983858	FUC	1	—
power = 3.0	0.377330	FUC	1	—
kappa = 0.9	0.633115	FUC	1	0.019727
kappa = 1.5	0.610544	FUC	1	0.018823
slope = 1.1	0.389769	FUC	1	0.012399
slope = 1.7	0.900312	FUC	1	—
r0 = 0.3	0.576151	FUC	1	0.016604
r0 = 0.9	0.657317	FUC	1	0.021211
nu = 1.0	0.625140	FUC	1	0.019290
nu = 3.0	0.625140	FUC	1	0.019290
persistence = 0.6	0.580915	FUC	1	0.022693
persistence = 1.2	0.643989	FUC	1	0.015249
dispersion = 0.8	0.633394	FUC	1	0.019744
dispersion = 1.2	0.603774	FUC	1	0.018567
utility delta = -0.2	0.660211	FUC	1	0.020390
utility delta = 0.2	0.585697	FUC	1	0.018221
utility delta = 0.8	0.481061	FUC	1	0.015197
fill delta = -0.015	0.625140	UC	1	—
fill delta = 0.015	infeasible	—	—	—
stock step = 1.0	0.625140	FUC	1	0.019290
stock step = 0.25	0.625140	FUC	1	0.019290
lower variance prob	0.633115	FUC	1	0.019727
higher variance prob	0.610544	FUC	1	0.018823
higher service with cost slack	infeasible	—	—	—
zero regime premium	0.000000	FUC	0	—

Grid gain uses the 0.1 accepted menu and the reoptimized continuous static comparator under the same reservations. For increased customer reservations the accepted static tier is reoptimized again. The full JSON records original-static feasibility separately. Active constraints are filled demand (F), customer utility (U), and physical cost (C), using a numerical tolerance of 10^{-7} . Lottery counts are reachable states randomized

by the returned LP policy; a positive count does not prove that every optimizer randomizes. Memory gain is the continuous full-state minus time-regime value, computed only when stock identification and the interior quadratic reduction are verified. Dashes do not assert zero. All robustness diagnostics are floating-point, not interval certificates.

EC.7.2 Training samples, derivative labels, and uncertainty

The graph critic uses degree twelve and thirteen fitted readout coefficients. Training seeds are 101 through 110. Each seed generates 96 graphs of size 32 or 64 and three scalar observations at each graph’s central and perturbed inherited-state vectors. Value-only training uses all 288 scalar observations. Value–gradient training uses the identical observations and adds their centered directional-difference residual to the fitting objective. It does not receive neighboring exact values unavailable to the comparator.

For the quadratic oracle and quadratic critic features, centered directional differences equal the exact directional derivative for every nonzero step. Consequently the derivative-weighted objective is also a finite-difference regularizer constructed from the same scalar information. Calling these two descriptions independent baselines would double count one method. The sensitivity file evaluates steps 0.01, 0.05, and 0.1 for the predesignated seed 101 on all four 1,024-service test graphs. The main step is 0.025. There is no teacher spatial grid or discretization-dependent differentiation rule in this experiment; the historical continuous-chain experiment retains that separate limitation.

Test graphs use the fixed seed 9091, four graphs per size, and coupling strengths one through four. Their topology is a cycle with an additional random matching. Tariff numerators are sampled from the integers 36 through 316 and divided by 100. The matrix is supplied in integer CSR form; the test arrays, deployed tier numerators, dual numerators, and fitted coefficients are retained. Training graphs and test graphs are distinct. The model is not retrained for 256 or 1,024 coordinates.

Table 4: Equal-information paired training-seed comparison

d	Mean improvement	Lower 95%	Upper 95%	Favorable seeds
64	3.07e-09	-1.08e-09	7.22e-09	6/10
256	1.76e-09	-1.53e-09	5.04e-09	6/10
1024	1.58e-09	-3.93e-09	7.08e-09	6/10

Improvement is value-only loss minus value–gradient loss per service. The Student intervals summarize variation across ten independent training seeds conditional on the four selected test graphs; they are not population guarantees across arbitrary operations networks. The unit is a training seed, not forty independent replications. Every interval includes zero.

The paired uncertainty calculation averages the four test graphs within a training seed before computing a Student interval across the ten seeds. It does not count all forty deployments as independent training replications. All three intervals contain zero. The result supports a controlled implementation comparison, not a broad claim of statistically established gradient-supervision superiority.

Table 5: All coupled portfolio sizes

d	Method	Mean loss/service	Max. bound/service	Time (ms)
64	Conjugate gradient	7.50e-10	4.88e-09	0.463
64	Analytic Chebyshev	2.57e-08	3.49e-07	0.206
64	Dense Cholesky	0.00e+00	5.61e-12	0.144
64	Sparse direct	0.00e+00	5.61e-12	0.226
64	Learned value–gradient	1.27e-08	3.64e-07	0.186
64	Learned value only	1.58e-08	5.38e-07	0.188
256	Conjugate gradient	8.62e-10	4.92e-09	0.676
256	Analytic Chebyshev	2.75e-08	3.82e-07	0.428
256	Dense Cholesky	0.00e+00	7.25e-12	0.775
256	Sparse direct	0.00e+00	7.25e-12	0.805
256	Learned value–gradient	1.50e-08	5.79e-07	0.409
256	Learned value only	1.68e-08	7.45e-07	0.412
1024	Conjugate gradient	6.11e-10	3.46e-09	1.524
1024	Analytic Chebyshev	2.69e-08	3.67e-07	1.220
1024	Dense Cholesky	0.00e+00	7.80e-12	18.010
1024	Sparse direct	0.00e+00	7.80e-12	7.619
1024	Learned value–gradient	1.52e-08	8.04e-07	1.214
1024	Learned value only	1.68e-08	9.01e-07	1.222

Same definitions and protocol as main Table 2. No favorable-seed selection is used.

EC.7.3 Arithmetic and timing scope

Timing uses one CPU environment with BLAS and OpenMP thread counts set to one. Each recorded time is the median of three calls containing the two required linear solves or graph-filter applications and customer-budget calibration. Offline sample generation and fitting, graph construction, and the rational post-deployment audit are excluded from that deployment time. Average value–gradient sample generation plus training is approximately 0.087 seconds in this environment. Hardware timings are observations, not complexity guarantees.

The learned readout stores thirteen float64 coefficients, or 104 bytes, in addition to the graph and the working vectors needed by every method. A dense matrix alone at 1,024 coordinates stores 8 MiB in float64; this is an accounting comparison, not a claim that a sparse direct solver uses a dense matrix. The graph and its factorization are not free. Sparse direct, dense Cholesky, conjugate gradients with relative tolerance 10^{-4} , and a fixed analytic degree-twelve Chebyshev inverse use the same deployment graph, tariffs, and acceptance calibration.

Each tier and multiplier is rounded to denominator 10^{10} . A downward correction ensures the payment inequality using exact fractions, rather than a floating tolerance. The independent checker confirms matrix symmetry, nonpositive off-diagonal entries, and a row sum of 21 for the integer matrix. Thus the matrix divided by ten is bounded below by $2.1I$. It recomputes the residual norm and dual slack with arbitrary-precision integer arithmetic. Stored reward and bound fractions must match exactly. The certification concerns the deployed rounded policy and the continuous-action quadratic problem; it neither treats floating

learned weights as exact real numbers nor certifies a general continuum HJB.

EC.8 Result-specific novelty and evidence map

The stock-support lemma and accepted quadratic reduction establish an economic identification result. Convexity and Riccati algebra are standard; the new assertion is that the same physical commitments force a policy reduction and permit an exact accepted information decomposition. The local perturbation theorem supplies primitive conditions for a customer-neutral improvement. The continuation program and its feasible rational policy quantify the cost of allowing customer exit.

The lattice theorem uses standard monotone-selection machinery under contract-specific compatibility. The ordered envelope exploits a known one-dimensional upper-hull construction; the quadratic grid theorem has its stated contract-discretization scope. Multi-tier inaction intervals use every competing action. The Bregman formula is a dual-coordinate identity, not a fast multidimensional hull theorem. The occupation formulation and Lagrangian upper bound use standard constrained-control duality; their role is independent certification of the economic comparison.

The learned graph critic combines an established polynomial representation with compatible value derivatives and acceptance calibration. Its architecture and least-squares readout are not claimed as new general-purpose learning algorithms. The new evaluation is equal in raw information, transfers across service dimensions, and supplies exact post-deployment loss bounds. The old two-seed chain study, manufactured diffusion checks, and least-squares conditioning parity remain separate evidence classes in the following historical sections.

EC.9 Nonlinear study: reproducibility and audit details

The executable entry point is `revisions/or-r10-20260921/reproduce.sh`. The verification-only entry point is `verify.py`; it reads serialized rational inputs and does not import the optimizer or the neural feature code. Training, deployment, raw timing, structural-test, and static-gate records are separate files. Every deployment includes its full rational tier vector, nonnegative multiplier vector, exact value, exact static comparator value, and exact upper-gap fraction. Every training endpoint includes an analogous record. The former R7 accepted hierarchy, customer-exit, and quadratic portfolio records are separately regenerated and replayed; they are not relabeled as nonlinear results.

The binary tree has transition probability one half and discount factor nine tenths. Maintenance curvature is three halves, temporal adjustment is three fifths, and each service tariff cycles through one, five quarters, and three halves. Training uses quartic coefficient one and graph strength one quarter. Cycle and random-edge tests use those same coefficients; geometric tests use coefficients two and one half; weighted-block tests use three and one. The last graph has within-block connection probability seven tenths, cross-block probability twelve hundredths, and integer edge weights from one to four. Connectivity is not imposed or required for strong convexity. Primitive resource shocks are bounded rational numbers and centered with the actual discounted occupation weights.

The learned potential applies softplus nonlinearities at three scales and seven shifts to each of three local transforms: identity, one smoothing operation, and two smoothing operations. Its 63 readout coefficients are learned by regularized least squares. Both objectives receive the same 72 scalar oracle values per training seed. The derivative-weighted objective adds 24 central-difference equations computed from those same values. The fixed ridge coefficient is one hundred-thousandth. Ten training seeds and two problem-generation seeds per family are disclosed in code. No fastest seed, favorable instance, or failed transfer is selected for reporting.

Nonlinear Newton steps use the exact quartic Hessian. The participation dual is optimized with non-negative multipliers; the dual’s floating-point exit status is retained only as a diagnostic. Newton-PCG precomputes one shifted baseline Hessian factorization per graph-tree instance and reuses it across changing Newton systems and all 15 repeated solves. Thus the direct method’s changing factorization cost is compared with an actual cached alternative, rather than treating repeated factorization as the only classical baseline. Setup is recorded separately; feature construction, acceptance repair, multiplier proposal, and fraction-based audit are included in the online candidate timings. Training time is separately retained and is not amortized into a claimed break-even point.

The exact repair first clips tiers to the compact box and rounds downward to a grid with denominator one hundred million. Because tariff entries are positive, multiplying all nonnegative tiers by the smallest subtree-budget ratio preserves the box and makes every payment inequality feasible. A second downward rounding preserves feasibility. The repair is deliberately simple, not claimed to be the closest feasible projection; its economic distortion is visible in the raw rewards. Consequently the static gate compares the repaired proposal with the exactly optimized static protocol before deployment.

The replay checker reconstructs the entire polynomial gradient from the recorded primitives, including both appearances of every temporal edge and both endpoints of every spatial edge. It reconstructs each ancestor multiplier contribution independently. No numerical tolerance is used for acceptance, multiplier signs, objective equality, or equality of the stored rational certificate. Structural checks additionally compare 120 edge-criterion cases with normalized linear programs, verify the tree summation identity with fractions, check 12 absolute-friction cases, and replay the zero-friction promise example with gap one thirty-second. These finite checks exercise the theorem implementation; the proofs do not rest on those tests.

A preliminary unanchored-readout run prompted the explicit reporting of severe-shift failures and the exact static gate. The final fitting architecture, sampling seeds, and regularization were not tuned to remove those failures. The final dataset differs from the preliminary calculation in replacing unchecked floating oracle values with feasible rationally certified values and explicit endpoint error bounds. Both value-only and derivative-weighted objectives receive this same change. All final negative cases remain disclosed; statistical inference concerns ten paired training-seed means, not a fictitious sample of 160 independent fitted networks.

EC.9.1 Cached-factor re-audit of the quadratic diagnostic

The unchanged R7 graph instances and the prespecified first learned-gradient training seed are replayed with cached sparse LU, cached dense Cholesky, Jacobi-preconditioned conjugate gradients, the analytic

polynomial filter, and a fresh sparse solve. For each method, the two right-hand sides used to calibrate acceptance are solved together where supported. Setup is excluded from online time but reported separately. Each timing includes the same exact rational certificate; all 72 final decisions and all repeated observations are retained. Table 6 replaces any interpretation of the old per-call factorization comparison as a fair amortized speedup result. The new comparison is still not at matched certified accuracy, and no universal learned advantage is asserted.

EC.9.2 Operator shift diagnostics

Table 7 reports the extremal eigenvalues of the discounted-weight-normalized Hessian at the static tier, as well as maximum weighted spatial degree. The training-sized spectrum is not used as a purported enclosure for shifted graphs. These floating diagnostics are distinct from the exact, dimension-independent maintenance-curvature lower bound used in each residual certificate.

Table 6: R10 re-audit of the unchanged quadratic portfolio task. End-to-end time includes multiplier calibration, rational repair, and exact audit. Setup is paid once per graph; four graph-specific medians are averaged, each using 15 repetitions after one warmup. The learned coefficients use the prespecified first training seed. Accuracy is not matched.

Services	Method	Setup (ms)	End-to-end (ms)	Max. bound/service
64	Sparse, fresh	0.00	1.10	5.61e-12
64	Sparse, cached	0.21	0.96	5.61e-12
64	Dense, cached	0.14	0.98	5.61e-12
64	Jacobi-PCG	0.10	3.00	6.03e-12
64	Analytic filter	0.00	1.13	3.49e-07
64	Learned gradient	0.00	1.11	1.31e-07
256	Sparse, fresh	0.00	2.46	7.25e-12
256	Sparse, cached	0.61	1.90	7.25e-12
256	Dense, cached	0.72	2.06	7.25e-12
256	Jacobi-PCG	0.10	4.15	7.57e-12
256	Analytic filter	0.00	2.04	3.82e-07
256	Learned gradient	0.00	2.02	5.79e-07
1024	Sparse, fresh	0.00	12.41	7.8e-12
1024	Sparse, cached	7.35	5.33	7.8e-12
1024	Dense, cached	12.92	14.13	7.8e-12
1024	Jacobi-PCG	0.18	7.86	7.64e-12
1024	Analytic filter	0.00	5.49	3.67e-07
1024	Learned gradient	0.00	5.47	3.42e-07

Table 7: Measured spectrum of the discounted-weight-normalized Hessian at the static tier and maximum weighted spatial degree. These floating diagnostics expose distribution shift; exact certificates use the proved lower curvature bound instead.

Instance	Coordinates	Minimum	Maximum	Weighted degree
training-template	28	6.26	11.71	2
cycle-4-3-33001	28	6.26	11.71	2
cycle-4-3-33002	28	6.26	11.71	2
erdos-8-3-33001	56	6.26	12.92	5
erdos-8-3-33002	56	6.26	13.35	6
geo-8-4-33001	120	9.17	19.63	5
geo-8-4-33002	120	9.17	19.41	5
block-32-5-33001	992	12.12	112.72	43
block-32-5-33002	992	12.12	128.05	46

References

- Abbasi B, Bruzda J, Gavirneni S (2022) Optimal operational service levels in vendor managed inventory contracts—an exact approach. *Operations Research Letters* 50(5):610–617. doi:10.1016/j.orl.2022.08.007.
- Boyd S, Vandenberghe L (2004) *Convex Optimization* (Cambridge University Press, Cambridge).
- Caggiano KE, Jackson PL, Muckstadt JA, Rappold JA (2007) Optimizing service parts inventory in a multiechelon, multi-item supply chain with time-based customer service-level agreements. *Operations Research* 55(2):303–318. doi:10.1287/opre.1060.0345.
- Czarnecki WM, Osindero S, Jaderberg M, Swirszcz G, Pascanu R (2017) Sobolev training for neural networks. *Advances in Neural Information Processing Systems* 30. arXiv:1706.04859.
- Defferrard M, Bresson X, Vandergheynst P (2016) Convolutional neural networks on graphs with fast localized spectral filtering. *Advances in Neural Information Processing Systems* 29.
- Federgruen A, Heching A (1999) Combined pricing and inventory control under uncertainty. *Operations Research* 47(3):454–475.
- Han J, Jentzen A, E W (2018) Solving high-dimensional partial differential equations using deep learning. *Proceedings of the National Academy of Sciences* 115(34):8505–8510.
- Henig M, Gerchak Y, Ernst R, Pyke DF (1997) An inventory model embedded in designing a supply contract. *Management Science* 43(2):184–189. doi:10.1287/mnsc.43.2.184.
- Hipp SK, Holzbaur UD (1988) Decision processes with monotone hysteretic policies. *Operations Research* 36(4):585–588. doi:10.1287/opre.36.4.585.
- Hosseiniifard Z, Shao L, Talluri S (2022) Service-level agreement with dynamic inventory policy: The effect of the performance review period and the incentive structure. *Decision Sciences* 53:802–826. doi:10.1111/deci.12506.
- Katok E, Thomas D, Davis A (2008) Inventory service-level agreements as coordination mechanisms: The effect of review periods. *Manufacturing & Service Operations Management* 10(4):609–624. doi:10.1287/msom.1070.0188.
- Kushner HJ, Dupuis P (2001) *Numerical Methods for Stochastic Control Problems in Continuous Time*, 2nd ed. (Springer, New York).
- Liang L, Atkins D (2013) Designing service level agreements for inventory management. *Production and Operations Management* 22(5):1103–1117. doi:10.1111/poms.12033.
- Ma J, Protter P, Yong J (1994) Solving forward-backward stochastic differential equations explicitly—a four step scheme. *Probability Theory and Related Fields* 98:339–359.

- Nasser S, Turcic D (2019) Temporary contract adjustment to a retailer with a private demand forecast. *Management Science* 65(1):209–229. doi:10.1287/mnsc.2017.2942.
- Plambeck EL, Zenios SA (2003) Incentive efficient control of a make-to-stock production system. *Operations Research* 51(3):371–386. doi:10.1287/opre.51.3.371.14954.
- Sieke MA, Seifert RW, Thonemann UW (2012) Designing service level contracts for supply chain coordination. *Production and Operations Management* 21(4):698–714. doi:10.1111/j.1937-5956.2011.01301.x.
- Topkis DM (1978) Minimizing a submodular function on a lattice. *Operations Research* 26(2):305–321.